



Cardiac Catheterization

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Cardiac catheterization is the passage of a catheter through peripheral arteries or veins into cardiac chambers, the pulmonary artery, and coronary arteries and veins.

Cardiac catheterization can be used to perform various tests, including:

- Intracardiac and intravascular pressure measurement (including ventricular filling pressures and pulmonary artery pressure)
- Angiography
- Detection and quantification of shunts
- Measurement of cardiac output (CO)
- Measurements of myocardial metabolism
- Endomyocardial biopsy
- Intravascular ultrasound (IVUS)

These tests define coronary artery anatomy, cardiac anatomy, cardiac function, and pulmonary arterial hemodynamics to establish diagnoses and help clinicians select treatment.

Cardiac catheterization is also the basis for therapeutic interventions (see [Percutaneous coronary intervention in Treatment of Coronary Artery Disease](#), [Treatment of Aortic Stenosis](#), [Treatment of Mitral Regurgitation](#)).

Diagnostic and therapeutic cardiac catheterization for [congenital heart disease](#) is not discussed in detail here.

Procedure for Cardiac Catheterization

Many but not all preprocedure protocols require patients to fast for 4 to 6 hours before cardiac catheterization. Most patients do not require overnight hospitalization unless a therapeutic intervention is also performed. The procedure is typically performed with local anesthesia at the access site and light

or moderate intravenous systemic sedation that keeps patients comfortable but awake and able to participate throughout.

Left heart catheterization

Left heart catheterization is most commonly used to assess:

- Coronary artery anatomy and presence of coronary artery disease

Left heart catheterization is also used to assess:

- Aortic blood pressure
- Aortic valve function
- Left ventricular pressure and function
- Mitral valve function
- Systemic vascular resistance

Left heart catheterization is performed via femoral, subclavian, radial, or brachial artery puncture, with a catheter passed into the coronary artery ostia and/or across the aortic valve into the left ventricle (LV). Radial artery access is preferred for coronary angiography and intervention as it is more comfortable and carries a lower risk of hematoma or pseudoaneurysm or arteriovenous fistula formation when compared with femoral artery access (1).

Catheterization of the left atrium (LA) and LV is occasionally performed using transseptal perforation during right heart catheterization, particularly during [electrophysiologic studies](#) when access to the left atrium is needed and the foramen ovale is not patent.

In addition to coronary artery evaluation, hemodynamic left heart catheterization can be performed to obtain direct left atrial and left ventricular pressure measurements and to directly measure transaortic and transmitral pressure gradients.

Right heart catheterization

Right heart catheterization is commonly used to measure:

- Right atrial pressure
- Right ventricular pressure
- Pulmonary artery pressure
- [Pulmonary artery occlusion pressure](#) (PAOP—see figure [Diagram of the Cardiac Cycle](#))

The most frequent indications for right heart catheterization are to assess hemodynamics to guide therapy in shock of unknown etiology or type, to evaluate the need for cardiac transplantation or mechanical cardiac support (eg, a ventricular assist device) in patients with severe cardiogenic shock, for periprocedural risk stratification among people with severe cardiopulmonary comorbidities, and to diagnose pulmonary hypertension.

PAOP (also called pulmonary capillary wedge pressure) approximates left atrial and left ventricular end-diastolic pressure in most cases. In seriously ill patients, PAOP helps assess volume status (by way of left ventricular end-diastolic pressure as a proxy for left ventricular preload) and, with simultaneous measurements of cardiac output, can help guide therapy.

Right heart catheterization is also useful for assessing cardiac filling pressures, pulmonary vascular resistance, tricuspid or pulmonic valve function, intracardiac shunts, and right ventricular pressure.

Right heart pressure measurements may help in the diagnosis of [cardiomyopathy](#), [constrictive pericarditis](#), and [cardiac tamponade](#) when noninvasive testing is nondiagnostic, and it is an essential part of the assessment for cardiac transplantation or mechanical cardiac support (eg, use of a ventricular assist device).

The [procedure](#) is performed via femoral, subclavian, internal jugular, or antecubital vein puncture. A catheter is passed into the right atrium, through the tricuspid valve, into the right ventricle, and across the pulmonary valve into the pulmonary artery.

Selective catheterization of the coronary sinus can also be performed.

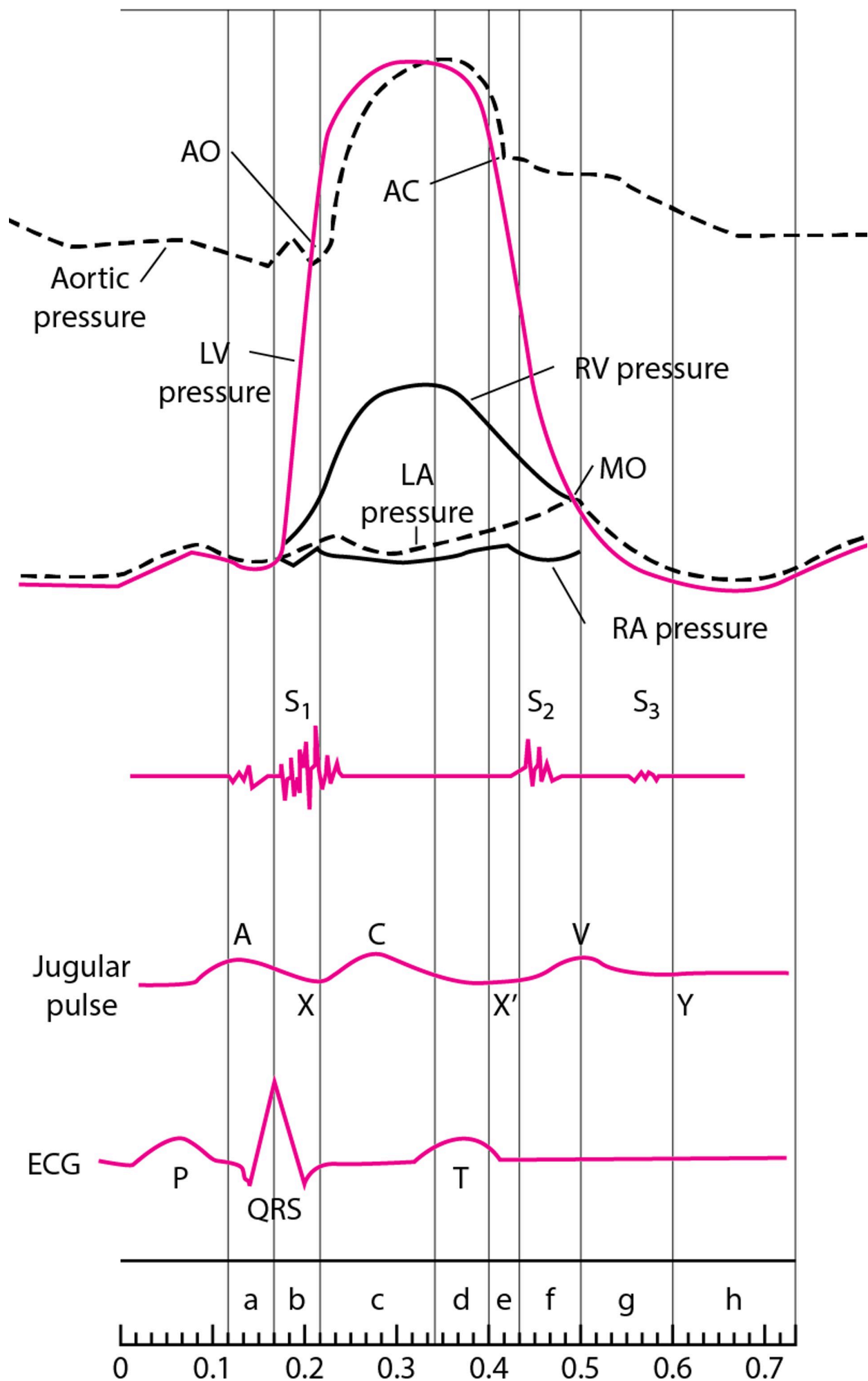
Hemodynamic assessment via right heart catheterization during exercise is increasingly being performed as part of the workup for dyspnea of uncertain etiology. The test can be performed at the same time as [cardiopulmonary exercise testing](#), called **invasive cardiopulmonary exercise testing**. This test is considered the gold standard for diagnosis of [heart failure with preserved ejection fraction](#) (HFpEF) as a cardiac limitation to exercise but is available at relatively few centers ([2](#)). Invasive cardiopulmonary exercise testing should be considered in patients at intermediate pretest probability for HFpEF if the diagnosis is uncertain after an initial evaluation. An increase in the PAOP > 25 mm Hg confirms the diagnosis when patients have signs and symptoms of heart failure irrespective of left ventricular ejection fraction.

Diagram of the Cardiac Cycle, Showing Pressure Curves of the Cardiac Chambers, Heart Sounds, Jugular Pulse Wave, and the ECG

The phases of the cardiac cycle are atrial systole (a), isovolumetric contraction (b), maximal ejection (c), reduced ejection (d), protodiastolic phase (e), isovolumetric relaxation (f), rapid inflow (g), and diastasis, or slow LV filling (h). For illustrative purposes, time intervals between valvular events have been modified, and the z point has been prolonged.

In the jugular venous pulse tracing, the A wave represents atrial contraction at end-diastole, C wave is the carotid artery impulse or bulging of the tricuspid valve into the right atrium in early systole, V wave is the increase in pressure and volume of right atrial filling in late systole and early diastole, X and X' descents represent movement of the lower part of the right atrium toward the ventricle with X' at end-systole, and the Y descent is tricuspid valve opening and ventricular filling in diastole.

AO = aortic valve opening; AC = aortic valve closing; LV = left ventricle; LA = left atrium; RV = right ventricle; RA = right atrium; MO = mitral valve opening.



Procedure references

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Specific Tests During Cardiac Catheterization

Angiography

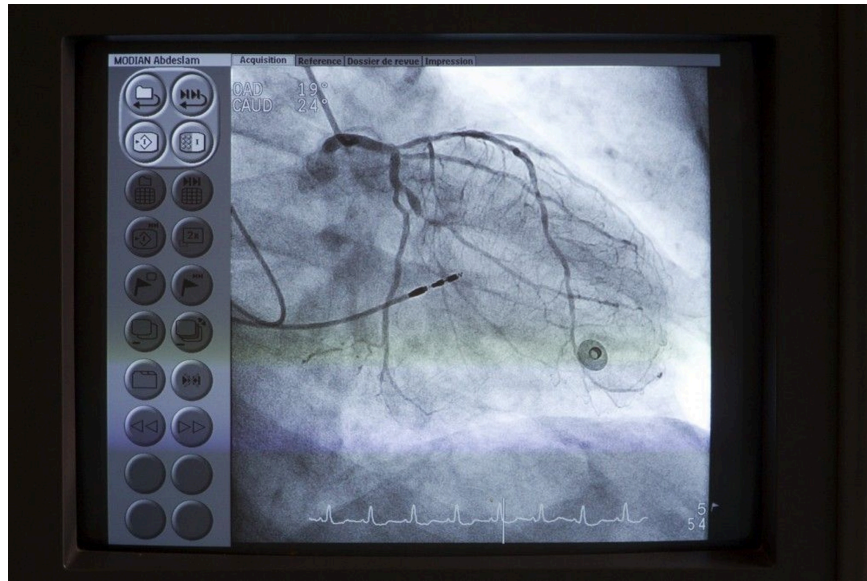
Injection of a radiopaque contrast agent into coronary or pulmonary arteries, the aorta, and cardiac chambers is useful in certain circumstances. Digital subtraction angiography, in which images of background structures are removed to improve visualization of the structures of interest, is used for nonmoving arteries and for chamber cineangiography.

Selective **coronary angiography** via left heart catheterization is used to evaluate coronary artery anatomy in various clinical situations, as in patients with suspected [coronary atherosclerotic](#) or [congenital coronary artery](#) disease, [valvular disorders](#) before valvular replacement, or unexplained [heart failure](#).

Aortic angiography via left heart catheterization is used to assess [aortic regurgitation](#), [coarctation](#), [patent ductus arteriosus](#), and [dissection](#).

Coronary Angiography

IMAGE



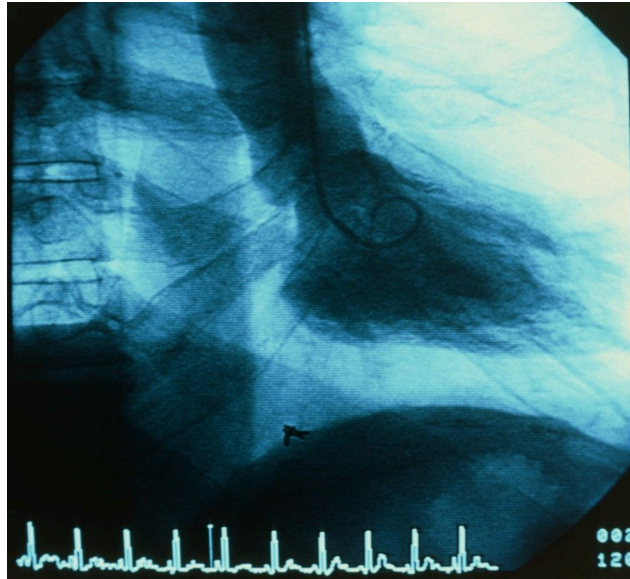
A catheter (upper left) is placed at the os of the left main coronary artery. Contrast is then injected into the coronary arteries under fluoroscopy to detect any anomalies such as stenoses.

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Pulmonary angiography via right heart catheterization can be used to diagnose [pulmonary embolism](#). Intraluminal filling defects or arterial cutoffs are diagnostic. Radiopaque contrast agent is usually selectively injected into one or both pulmonary arteries and their segments. Computed tomographic pulmonary angiography (CTPA) has largely replaced right heart catheterization for diagnosis of acute pulmonary embolism. Pulmonary angiography via right heart catheterization remains commonly used to determine a management plan for suspected chronic thromboembolic disease. Pulmonary angiography can also be used evaluate pulmonary vascular abnormalities such as branch pulmonary artery stenosis.

Ventriculography (LV Gram)

IMAGE



A catheter (center) is inserted into the left ventricle (LV). Contrast is then injected into the LV under fluoroscopy and the ejection fraction is visually estimated.

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Ventriculography, performed via right and/or left heart catheterization, is used to visualize ventricular wall motion and ventricular outflow tracts, including subvalvular, valvular, and supra-valvular regions. It is also used to estimate severity of [mitral valve regurgitation](#) and determine its pathophysiology. After left ventricular mass and volume are determined from single planar or biplanar ventricular angiograms, end-systolic and end-diastolic volumes and ejection fraction can be calculated.

Coronary artery flow measurements

Coronary angiography shows the presence and degree of stenosis but not the functional significance of the lesion (ie, how much blood flows across the stenosis) or whether a specific lesion is likely to be the cause of symptoms.

Extremely thin guidewires with pressure sensors or Doppler flow sensors can be used to estimate coronary artery blood flow, which is expressed as fractional flow reserve (FFR). FFR is the ratio of maximal flow through the stenotic area to normal maximal flow obtained during hyperemia (most commonly with adenosine); an FFR of < 0.75 to 0.8 is considered abnormal ([1](#)). Other techniques of measuring coronary blood flow, which do not require hyperemia induced by adenosine, are collectively termed non-hyperemic pressure ratios (NHPR). They include instantaneous wave-free ratio (iFR) and diastolic hyperemia-free ratio (DFR). NHPR techniques measure gradients across a stenosis during a period in diastole; an NHPR of ≤ 0.89 is considered abnormal ([2](#), [3](#), [4](#)).

These flow estimates correlate well with the need for intervention and long-term outcome; patients with lesions with FFR > 0.8 or NHPR > 0.89 do not seem to benefit from placement of a stent. These flow measurements are most useful with intermediate lesions (40 to 69% stenosis) and with multiple lesions

(to identify those that are clinically most significant), or with proximal lesion affecting large territories ([5](#), [6](#), [7](#)).

Noninvasive estimates of FFR can also be derived from data obtained during coronary CT angiography.

Intravascular ultrasound (IVUS)

Miniature ultrasound transducers on the end of coronary artery catheters can produce images of coronary vessel lumina and walls and delineate blood flow. Intravascular ultrasound is used to guide coronary angiography, including to guide optimal stent placement during percutaneous coronary intervention. It is also used to troubleshoot acute coronary complications and stent failure, to detect cardiac allograft vasculopathy after heart transplantation, to identify coronary artery dissections, and to guide endovascular therapy in aortic interventions ([8](#), [9](#), [10](#)).

Optical coherence tomography (OCT)

Optical coherence tomography is an optical analog of intracoronary ultrasound imaging that measures the amplitude of backscattered light to determine the temperature of coronary plaques and can help determine whether lesions are at high risk of future rupture (leading to [acute coronary syndromes](#)). Unlike in IVUS, injection of contrast is required to obtain images. The indications and appropriate use for OCT are uncertain.

Tests for cardiac shunts

Measuring blood oxygen content at successive levels in the heart and great vessels can help determine the presence, direction, and volume of central shunts. The maximal normal difference in oxygen content between structures is as follows:

- The pulmonary artery and right ventricle: 0.5 mL/dL (0.5 vol%)
- The right ventricle and right atrium: 0.9 mL/dL (0.9 vol%)
- The right atrium and superior vena cava: 1.9 mL/dL (1.9 vol%)

If the blood oxygen content in a chamber exceeds that of the more proximal chamber by more than these values, a left-to-right shunt at that level is probable. Right-to-left shunts are strongly suspected when left arterial, left ventricular, or arterial oxygen saturation is low ($\leq 92\%$) and does not improve when pure oxygen (fraction of inspired oxygen = 1.0) is given. Left heart or arterial desaturation plus increased oxygen content in blood samples drawn beyond the shunt site on the right side of circulation suggests a bidirectional shunt.

The overall ratio of pulmonary to systemic blood flow ($Q_p:Q_s$) can be also calculated using measured systemic arterial, mixed venous, and pulmonary arterial oxygen saturation, combined with an assumed or measured pulmonary venous oxygen saturation. This ratio can be used to estimate the direction and magnitude of shunting.

Measurement of cardiac output and flow

Cardiac output (CO) is the volume of blood ejected by the heart per minute (normal at rest: 4 to 8 L/minute). Techniques (see table [Cardiac Output Equations](#)) used to calculate CO include:

- Fick cardiac output technique
- Indicator-dilution technique
- Thermodilution technique

TABLE

Cardiac Output Equations

Fick technique

$$CO = \frac{\text{Ambient } O_2 - \text{Expired } O_2 \text{ (mL/min)}}{(1.36) (\text{Hb g/dL}) \times (\text{SaO}_2 - \text{SvO}_2)}$$

Numerator is oxygen consumption (O₂ absorbed by lungs in mL/minute).

Indicator-dilution technique

$$CO = \frac{\text{Injectate mass (mg)}}{\int_{\infty} C(t) dt}$$

Denominator is the sum of contrast agent concentrations (C) at each time interval (t).

Thermodilution technique

$$CO = \frac{(T_B - T_I) \times \text{injectate volume (mL)} \times 53.5}{\int_{\infty} T_B(t) dt}$$

T_B – T_I is the difference between body and injectate temperatures; injectate is usually [dextrose](#) or saline. Denominator is the sum of changes in temperature at each time interval (t).

SaO₂= arterial oxygen saturation (%); SvO₂= mixed venous oxygen saturation (%), measured in the pulmonary artery.

With the **Fick technique**, CO is proportional to oxygen consumption divided by arteriovenous oxygen difference.

Dilution techniques rely on the assumption that after an indicator is injected into the circulation, it appears and disappears proportionately to CO.

Usually, CO is expressed in relation to body surface area (BSA) as the cardiac index (CI) in L/minute/m² (ie, CI = CO/BSA—see table [Normal Values for Cardiac Index and Related Measures](#)). BSA can be estimated using DuBois height (ht)–weight (wt) equation:

$$BSA \text{ in } m^2 = (wt \text{ in } kg)^{0.425} \times (ht \text{ in } cm)^{0.725} \times 0.007184$$

TABLE

Normal Values for Cardiac Index and Related Measurements

Measurement	Normal Value	SD
Oxygen uptake	143 mL/minute/m ² *	14.3
Arteriovenous oxygen difference	4.1 dL	0.6
Cardiac index	3.5 L/minute/m ²	0.7
Stroke index	46 mL/beat/m ²	8.1
Total systemic resistance	1130 dynes-second-cm ⁻⁵	178
Total pulmonary resistance	205 dynes-second-cm ⁻⁵	51
Pulmonary arteriolar resistance	67 dynes-second-cm ⁻⁵	23

* Varies with body mass index.

SD = standard deviation.

Adapted from [Barratt-Boyes BG, Wood EH](#). Cardiac output and related measurements and pressure values in the right heart and associated vessels, together with an analysis of the hemodynamic response to the inhalation of high oxygen mixtures in healthy subjects. *J Lab Clin Med*. 1958;51(1):72–90.

CLINICAL CALCULATORS

[Cardiac Output MultiCalc](#)



CLINICAL CALCULATORS

[Cardiac Output](#)



Endomyocardial biopsy

Endomyocardial biopsy helps assess transplant rejection and myocardial disorders due to infection, inflammation, or infiltrative diseases. The biopsy catheter (bioptome) can be passed into either

CLINICAL CALCULATORS

[Body Surface Area \(Du Bois Method\)](#)



ventricle, usually the right. Three to 5 samples of myocardial tissue are removed from the septal endocardium. The main complication of endomyocardial biopsy, cardiac perforation, occurs in 0.2 to 0.7% of patients ([11](#), [12](#)); it may cause hemopericardium leading to [cardiac tamponade](#). Injury to the tricuspid valve and supporting chordae may also occur and can lead to [tricuspid regurgitation](#).

Pulmonary vasodilator testing

During a cardiac catheterization to evaluate for [pulmonary hypertension](#), pulmonary artery pressure, pulmonary vascular resistance, and cardiac output are measured at baseline, and then after oxygen and/or [nitric oxide](#) administered. The response (or lack thereof) to pulmonary vasodilators can guide pulmonary hypertension therapy.

Tests during cardiac catheterization references

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Contraindications to Cardiac Catheterization

Relative contraindications to cardiac catheterization include:

- [Acute kidney injury](#)
- [Chronic kidney disease](#)
- [Coagulopathy](#)
- Fever
- [Radiopaque contrast agent allergies](#) in patients who have not been appropriately premedicated
- Systemic infection
- Uncontrolled [arrhythmia](#)
- Uncontrolled [hypertension](#)
- Uncompensated [heart failure](#)

Relative contraindications balance the urgency of the procedure (eg, in an acute myocardial infarction vs an elective case) and the severity of the contraindicating disorder. Periprocedural management of anticoagulants or antiplatelet medications is individualized based on the type of procedure (ie, arterial vs venous access), the urgency of the procedure, the indication for the medication, and the patient's risk of bleeding. Catheterization laboratories frequently have policies for the periprocedural management of these medications.

Complications of Cardiac Catheterization

The incidence of major complications during or after **diagnostic cardiac catheterization** is approximately 1% ([1](#), [2](#), [3](#)).

Most complications are minor and can be easily treated. Serious complications of diagnostic catheterization (eg, [cardiac arrest](#), [anaphylactic reactions](#), [shock](#), [seizures](#), renal toxicity) are rare. Mortality rate after diagnostic catheterization is 0.01% to 0.7% ([1](#), [2](#), [3](#)), with not all deaths caused by the procedure. [Myocardial infarction](#) (< 0.011%) and [stroke](#) (0.03% to 0.17%) may result in significant morbidity.

In patients undergoing percutaneous coronary interventions (PCI), major complication rates are approximately 4.5%, generally higher than those for patients undergoing diagnostic catheterization ([2](#)). Reported mortality during PCI is 0.6 to 4.8% (higher in patients undergoing emergent intervention for ST-segment elevation myocardial infarction) ([4](#)). The approximate rate of periprocedural myocardial infarction is 2 to 18%, acute kidney injury is 4 to 7%, stroke is up to 1%, and major bleeding is 1 to 5% ([5](#)).

Patient factors that increase risk of complications include:

- [Chronic kidney disease](#)
- [Chronic obstructive pulmonary disease](#) (COPD)
- [Diabetes](#)
- [Heart failure](#)
- Increasing age
- [Peripheral artery disease](#)
- [Valvular heart disease](#)

Contrast agent complications

Injection of radiopaque contrast agent produces a transient sense of warmth throughout the body in many patients. Tachycardia, a slight fall in systemic pressure, an increase in cardiac output, nausea, vomiting, and coughing may occur. Rarely, bradycardia occurs when a large amount of a contrast agent is injected; asking the patient to cough often restores normal rhythm by increasing coronary and cerebral perfusion and clearing contrast from coronary arteries.

More serious reactions (see also [Radiographic Contrast Agents and Contrast Reactions](#)) include:

- Allergic-type contrast reactions
- Contrast-induced acute kidney injury

Allergic-type reactions to contrast include urticaria and conjunctivitis, which usually respond to [diphenhydramine](#) 50 mg IV. Severe hypersensitivity reactions to iodinated radiopaque contrast, such as [anaphylaxis](#) and [anaphylactoid reactions](#) with bronchospasm, laryngeal edema, and dyspnea, are rare reactions, with an approximate frequency of 0.02 to 0.04% ([6](#)). One study of patients receiving intra-arterial contrast injection specifically found that 3.6% of patients had an immediate hypersensitivity reaction and 15% had a delayed reaction; however, the rate of severe reactions was only 0.3% ([7](#)). These reactions should be treated immediately with [epinephrine](#). Patients with a history of allergic reaction to

contrast, particularly after prior intra-arterial injection, may be premedicated with oral [prednisone](#) (or parenteral [hydrocortisone](#) if the study is urgent) and [diphenhydramine](#).

[Contrast-induced acute kidney injury](#) is defined as impairment of renal function (a 25 to 50% increase in serum creatinine from baseline or a 0.3 to 0.5 mg/dL [23 to 44 micromole/L] increase in absolute value) within 2 to 7 days of IV contrast administration judged to be caused by contrast rather than alternative causes ([8](#), [9](#)). For patients at risk, using the lowest possible dose of low-osmolar or iso-osmolar contrast, discontinuing nephrotoxic medications, avoiding multiple contrast studies within a short period of time, and providing intravascular volume expansion before and after contrast exposure likely reduces this risk. Statins may reduce the risk of acute kidney injury. For some high-risk patients in whom the benefits of the procedure outweigh the risks, prophylactic dialysis may be performed.

Catheter-related complications

If the catheter tip contacts the ventricular endocardium, ventricular ectopy or arrhythmias commonly occur, but ventricular fibrillation is rare. If it occurs, [direct current cardioversion](#) (DC cardioversion) is administered immediately.

Disruption of an atherosclerotic plaque by the catheter can release a shower of atheroemboli. Emboli from the aorta may cause [stroke](#) or [nephropathy](#). Emboli from proximal to distal coronary arteries may cause [myocardial infarction](#).

Coronary artery dissection is also possible.

Access site complications

Access site complications include:

- Bleeding
- Hematoma
- Pseudoaneurysm
- Arteriovenous (AV) fistula
- Limb ischemia

Bleeding from the access site may occur and usually resolves with compression. Mild bruises and small hematomas are common and do not require specific investigation or treatment.

A large or enlarging lump should be investigated using ultrasound to distinguish hematoma from pseudoaneurysm. A bruit at the site (with or without pain) suggests an AV fistula, which can be diagnosed using ultrasound. Hematomas usually resolve with time and do not require specific therapy. Pseudoaneurysms and AV fistulas usually resolve with compression; those that persist may require surgical repair.

Radial artery access carries a much lower risk of hematoma or pseudoaneurysm or AV fistula formation when compared with femoral artery access ([10](#)).

Complications references

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More Information

The following English-language resource may be useful. Please note that The Manual is not responsible for the content of this resource.

[American College of Radiology \(ACR\) Manual on Contrast Media.](#) 2025 ACR Committee on Drugs and Contrast Media provides a guide for safe and effective use of contrast media

Drug Information for the Topic



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